Null Effects of Chronic Social Isolation on *Drosophila melanogaster* in Open-Field Assay

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Abstract

Social isolation is a known stressor with profound physiological effects, but its specific impact on anxiety-like behaviours in *Drosophila melanogaster* remains unresolved. While early-life isolation affects fly lifespan and sleep, its role in baseline emotional reactivity is poorly understood. To determine if social isolation induces anxiety-like states, we reared male flies in either social groups or complete isolation for six days post-eclosion. We then evaluated their behaviour using a 10 mm open-field assay, measuring thigmotaxis (wall-following), freezing, and locomotor vigour. We found no significant differences between group-housed and single-housed flies across any behavioural or kinematic metric. These findings indicate that six days of isolation does not sufficiently alter baseline arousal or spatial preference in an open-field paradigm, suggesting that while isolation acts as a metabolic stressor in flies, it may not manifest as traditional anxiety-like behaviour without the presence of acute secondary stressors.

Introduction

Social environments have a profound impact on animal physiology and behaviour. In mammalian models, social isolation is a well-documented stressor that routinely increases anxiety-like behaviours, often measured via increased wall-following in open-field testing (Mohammad et al., 2016). In *Drosophila melanogaster*, chronic social isolation acts as a stressor, but its primary signature appears to be metabolic and cellular rather than purely psychogenic. For instance, isolation induces cellular stress pathways, such as the unfolded protein response (Brown et al., 2017), and triggers a starvation-like metabolic state that significantly suppresses sleep (Li et al., 2021). Conversely, group housing, the standard "control" in many paradigms, paradoxically exerts its own deleterious effects. High-density social environments and interactions with older individuals can significantly reduce both stress resistance and overall lifespan in flies (Lin et al., 2022; Leech et al., 2017). This ambiguity makes it unclear whether single housing, group housing, or both act as primary behavioural stressors.

Therefore, this study aims to understand how early-life social isolation affects exploratory and anxiety-like behaviour in male *Drosophila*. We hypothesised that if isolation triggers psychogenic anxiety, flies will exhibit increased freezing and thigmotaxis. Alternatively, if the metabolic stress of isolation dominates, isolated flies may instead exhibit foraging-like hyperactivity.

Methods

Fly preparation Male *Drosophila melanogaster* flies were reared at 25°C on a 16:8-hour light/dark cycle. To manipulate early-life social experience, flies within 8 hours of eclosion were randomly assigned to one of two housing conditions for six days prior to behavioural testing: a social-housed Control group (reared in groups of 30 males per vial, n = 16) and a single-housed Experimental group (isolated individually per vial, n = 17). A duration of six days was selected as it represents the threshold for chronic social isolation in *Drosophila*, which is known to induce significant metabolic and behavioural shifts compared to acute isolation (Li et al., 2021).

Open-Field Behavioural Assay and Video Acquisition Adult baseline locomotion and anxiety-like behaviours were evaluated using a custom-built open-field setup. Individual flies were introduced into circular arenas measuring 10.0 mm in diameter (Mohammad et al., 2016). This small-diameter paradigm was chosen to minimize the initial exploratory activity component and emphasize the measurement of "stimulated activity" induced by the novel environment (Lueningschroer-Wang et al., 2025; Soibam et al., 2012). The restricted ceiling height (3.18 mm) was chosen to prevent vertical flight and jumping, ensuring strict two-dimensional tracking (Valente et al., 2007).

Following a two-minute acclimation period, video recordings were captured from above for a continuous 10-minute observation period, a standard duration for capturing stable *Drosophila* open-field metrics (Chiu et al., 2021; Mohammad et al., 2016; Lueningschroer-Wang et al., 2025). The optical setup was standardized to maintain consistent spatial calibration wherein the focal plane of the camera was displaced 111.7 mm from the assay surface (Valente et al., 2007).

Kinematic Extraction and Statistical Analysis A custom Python script was used to track the displacement of the fly centre from the centre of the open field arena (see technical appendix for further information). Several locomotor and behavioural metrics were quantified.

Behavioural metrics were aggregated and averaged across the full 10-minute trial for each replicate. Comparisons between the social-housed and single-housed groups were conducted using independent samples t-tests assuming heteroskedasticity. All statistical outcomes were evaluated at a significance threshold of $\alpha = 0.05$.

Measure	Description	Control	Experimental	p-value
Distance	mm, total path distance over the test period (Mohammad et al., 2016)	2790 ± 670	3140 ± 563	0.170
Thigmotaxis	Percentage, time spent in the outer boundary zone ($r > 3.5$mm) (Soibam et al., 2012)	70.5 ± 14.3	67.2 ± 17.3	0.716
Freezing	Percentage, time instantaneous speed $v < 2.0$mm/s (Chiu et al., 2021)	23.9 ± 11.5	20.4 ± 8.7	0.285
Meandering	degrees/mm, absolute change in heading angle over distance travelled (Valente et al., 2007)	158 ± 31	158 ± 31	0.982
Centre entries	Counts, crossings from the boundary zone into the centre zone ($r < 3.5$ mm) (Lueningschroer-Wang et al., 2025)	164 ± 86	227 ± 136	0.043

Table 1: Summary of locomotor and anxiety-like behavioural metrics in the open-field assay. Values represent the sample mean ± SD for socially-housed controls (n=16) and single-housed experimental flies (n=17). P-values were calculated using independent samples t-tests assuming heteroskedasticity.

Results

Total distance travelled did not differ between the social-housed control and single-housed experimental groups (p=0.170). Primary measures of fear-like defensive states remained consistent regardless of housing; groups showed no significant difference in thigmotaxis (p=0.716) or freezing time (p=0.285). Path efficiency, measured via mean meandering, was also statistically indistinguishable (p=0.982). While raw data initially indicated isolated flies entered the arena centre more frequently (p=0.043), normalizing centre entries by total path distance rendered this difference non-significant (p=0.114), indicating an artefact of movement variation rather than a true spatial preference. Overall, six days of isolation is insufficient to alter baseline arousal or valence states typically captured in the open-field paradigm.

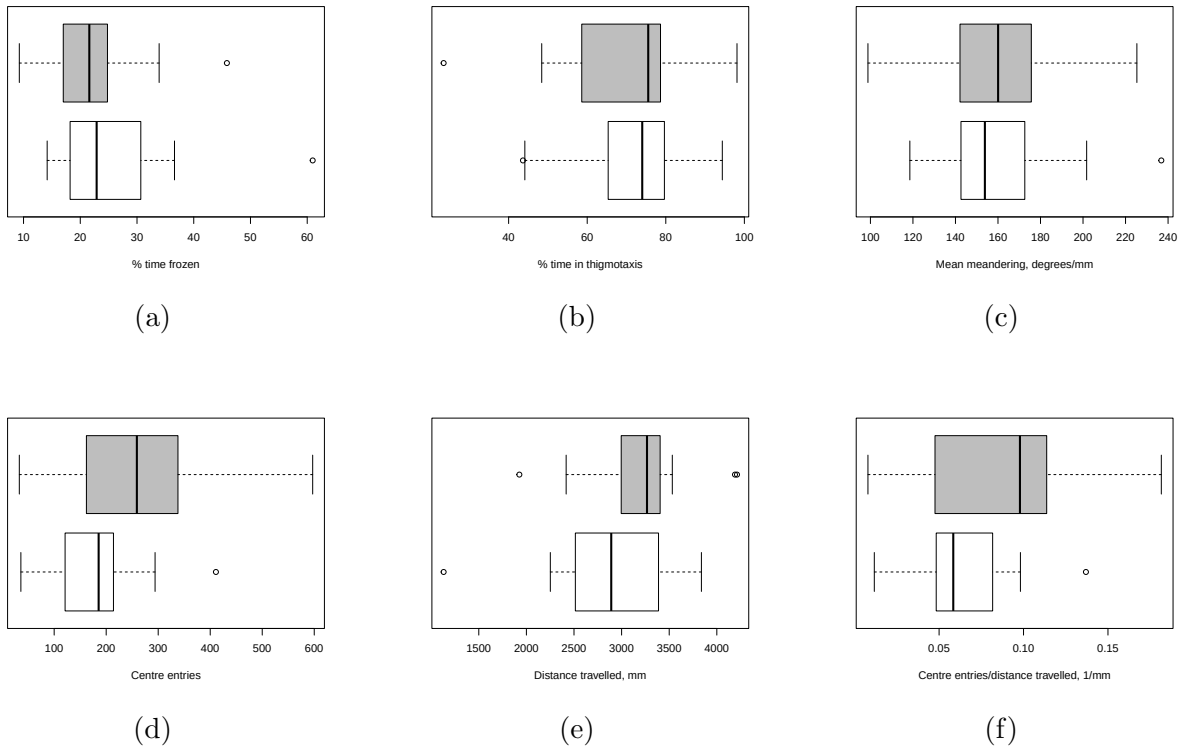

Figure 1: Effects of early-life housing conditions on open-field behaviours in adult *Drosophila*. Box plots display the median (centre line) and interquartile range (box) for (a) time frozen, (b) thigmotaxis, (c) mean meandering, (d) path distance travelled, (e) total centre entries, and (f) centre entries normalised by distance. Whiskers indicate the full range of the data excluding outliers. Control flies are shown in white and isolated flies in grey. No significant differences were observed across any metric.

Discussion

Our findings indicate that six days of chronic social isolation does not significantly alter baseline anxiety-like behaviour in an open-field assay. Contrary to our hypothesis, isolated flies did not exhibit heightened defensive states; thigmotaxis and freezing behaviours were statistically indistinguishable from socially housed controls.

While our primary spatial anxiety metrics yielded null results, our kinematic data highlighted a non-significant but biologically intriguing trend in locomotor vigour. The isolated experimental group demonstrated a higher average total path distance (3140 mm) compared to controls (2790 mm). Although our study was likely underpowered to detect a statistically significant difference in this metric due to massive individual variation (p=0.170), this trend aligns with recent neurobiological findings. Chronic isola-

tion in Drosophila induces a starvation-like state (Li et al., 2021); therefore, this increased path distance may not represent "anxiety," but rather a state of heightened foraging vigour and exploratory hyperactivity triggered by the metabolic illusion of starvation. Future research utilizing larger sample sizes should prioritize path distance and kinematic vigour as primary metrics of isolation-induced stress.

More importantly, these null findings emphasize the ambiguous nature of social stress in Drosophila and necessitate a critical re-evaluation of the assays used to measure it. In rodents, isolation directly triggers psychological distress that manifests as anxiogenic spatial avoidance. In flies, however, our results suggest that the "stress" of isolation does not inherently map onto the "emotion primitives" of fear or anxiety measured by the standard open-field test (Lueningschroer-Wang et al., 2025). Furthermore, because group housing can induce competitive stress and reduce lifespan (Lin et al., 2022; Leech et al., 2017), our "control" social housing may not represent a relaxed baseline state, but rather a different type of stressor entirely. We may be comparing starvation-stress (isolation) against social-stress (group housing), both of which might suppress baseline open-field anxiety in favour of other survival behaviours.

While we chose a 10 mm open field assay to align with previous literature, larger arenas could better isolate spatial preference in future studies (E. Hoopfer, personal communication, March 3, 2026). In accordance with the body size to arena ratio of mammals frequently tested such as rats, a larger arena would make the distinction between a stressful centre-exploratory behaviour and thigmotaxis clearer, although it would present challenges with resources and camera distortion in video tracking.

To properly capture an isolation-induced anxiety phenotype, future open-field paradigms may need to pivot from assessing baseline states to assessing reactive states. It is highly plausible that social isolation acts as a primer that sensitizes the nervous system, rather than a direct trigger for spatial anxiety. Future iterations of this experiment should employ a "two-hit" stress model, applying an acute, secondary stressor, such as a mechanical shock or sudden heat pulse, immediately prior to open-field testing. This approach could reveal whether isolated flies possess a hyper-reactive emotional state that unperturbed, baseline open-field assays fundamentally fail to detect.

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Technical appendix

Foreground extraction To isolate the movement of individual Drosophila against the stationary arena, video processing was conducted using a custom Python script utilising the OpenCV library. We adopted a background subtraction methodology inspired by the Ctrax project (Branson et al., 2009). The script constructs a robust background plate by randomly sampling frames across the video and calculating the median pixel intensity for each spatial coordinate.

During frame-by-frame analysis, the current frame is subtracted from this median background model. This produces a difference image where stationary background noise is eliminated, leaving only the moving subject.

Contour Detection and Centroid Tracking The difference image is subsequently converted to grayscale and subjected to binary thresholding to create a high-contrast mask of the fly. Topological contours of the resulting binary blob are identified. To pinpoint the exact location of the fly, the script calculates the image moments of the largest contour to determine its geometric centre (centroid). This yields raw pixel coordinates (x, y) for the subject at each frame (captured at 30 frames per second, $\Delta t \approx 0.033$s).

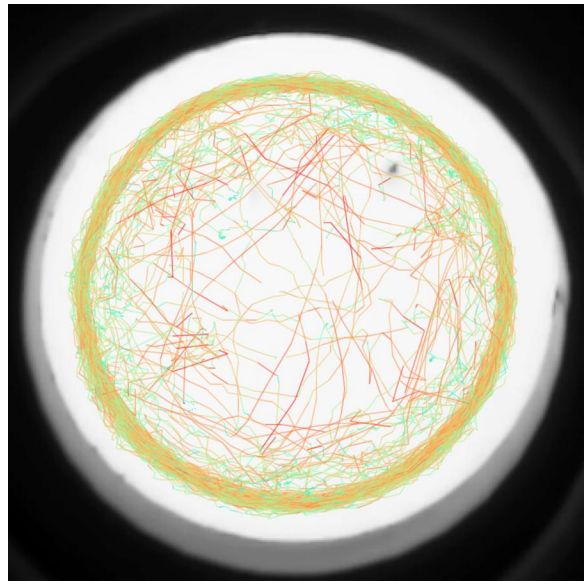

Figure 2: An example of a trial, with the tracked path traced and superimposed on computed background plate. The colour is logarithmically related to the computed instantaneous speed at the given point (blue: ∼0.0 mm/s; red: 40.0+ mm/s).

Spatial Calibration and Coordinate Processing To convert pixel coordinates into standard metric units (mm), a scaling factor was established based on the known 10.0 mm diameter of the physical testing arenas. The arena boundaries were detected using Hough Circle Transform algorithms, identifying the geometric centre (x_c, y_c) and the pixel radius of each chamber.

Crucially, to maintain the fidelity of micro-behaviours (e.g., postural shifts, hesitations), no artificial "jitter" filtering, coordinate snapping, or rolling-average spatial smoothing was applied to the raw trajectories. Furthermore, there was no ceiling filtering for erroneous detections of instantaneous speed, and each concern-

ing case was inspected manually to verify proper tracking.

Kinematic Calculations Once raw metric coordinates were established, a secondary Python class leveraging the *pandas* library was used to calculate continuous kinematic variables:

- **Instantaneous Speed** Calculated as the Euclidean distance travelled between consecutive frames divided by the time delta:

$$v = \frac{\sqrt{(x_t - x_{t-1})^2 + (y_t - t_{t-1})^2}}{\Delta t}$$

- **Radial Distance (r)** The fly's distance from the geometric centre of the arena:

$$r = \sqrt{(x_t - x_c)^2 + (y_t - y_c)^2}$$

Derivation of Behavioural Metrics To quantify complex emotional states and spatial decision-making, the pipeline derived several specialized metrics from the base kinematics:

- **Thigmotaxis and Centre Zones** A binary mask was applied based on radial distance. Frames where $r > 3.5$ mm were flagged as edge/thigmotaxis behaviour, while frames where $r \leq 3.5$ mm were flagged as centre-zone behaviour. Centre entry events were counted by detecting Boolean shifts from edge to centre across consecutive frames.

- **Freezing and Stop Bouts** A velocity threshold of 2.0 mm/s was established. Time spent below this threshold was classified as freezing. Stop bouts were enumerated by detecting transitions from an active state ($v \geq 2.0$ mm/s) to a frozen state.

- **Path Meandering** To measure path tortuosity, the change in heading angle ($\Delta\theta$) was calculated. To elegantly handle 360-degree wrap-around, the differential angle was computed as:

$$d\theta = |\arctan(\sin(\Delta\theta), \cos(\Delta\theta))|$$

Meandering was then defined as $d\theta$ divided by the step distance. To prevent mathematical explosions from stationary sub-pixel jitter, meandering was strictly calculated only when the fly was actively locomoting ($v \geq 2.0$ mm/s), and extreme outliers exceeding the 95th percentile were excluded.

Data Aggregation Finally, the script aggregated these high-resolution frame-by-frame calculations into functional replicate summaries (Mean Speed, Percentage of Time Freezing, Centre Entry Count, etc.) across defined time phases, exporting a master, wide-format .csv file for statistical analysis.

The code is available on GitHub. I would like to extend my gratitude toward Professor Eric D. Hoopfer for his help in developing the detection scripts as well as the data analyses.